Student Number

NORTH SYDNEY GIRLS HIGH SCHOOL

2015

Higher School Certificate

Trial Examination

Physics

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- **Write your student number at the top of this paper and the multiple choice answer sheet.**

Total Marks – 100

Section I – Core (90 Marks)

Part A – *20 Marks*

- **Attempt questions 1- 20**
- Allow about 35 minutes for this part

Part B – *70 Marks*

- **Attempt questions 21 - 35**
- Allow about 2 hour and 5 mins for this part

Section II – Option (10 Marks)

- **Attempt question 36**
- Attempt all questions in this section
- Allow about 20 minutes
- Answer in the space provided.

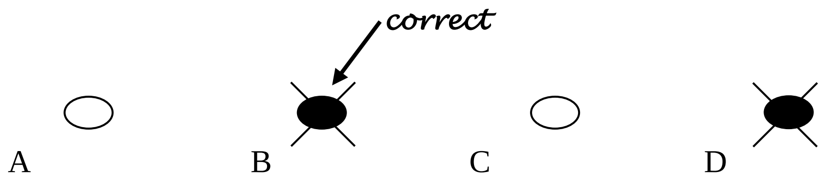
Part A Instructions:

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: 2 + 4 = (A) 2 (B) 6 (C) 8 (D) 9

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word **correct** and drawing an arrow as follows:



Part B Instructions:

- Questions are of different mark value.
- Students should allocate their time according to the mark value of each question.
- Answer all questions in the spaces provided.

Section I - Part A

Multiple Choice: Questions 1 – 20 (1 mark each)
Put your answers on the Multiple Choice answer sheet provided
Allow about 35 minutes for this part

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1. A space ship with an astronaut is in Earth orbit. Which of the following statements is correct?

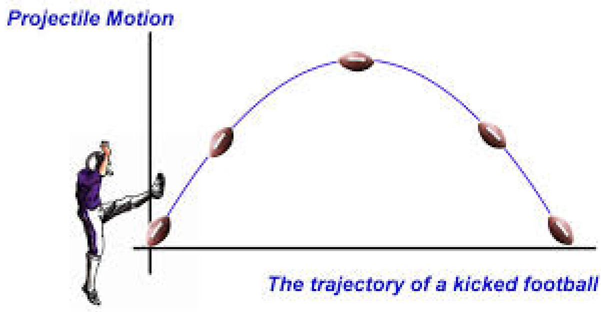
    (A) The astronaut notices that time on the ship is moving slower relative to the ground.

    (B) The astronaut notices that time on the ground is moving slower relative to herself.

    (C) The astronaut does not notice any time rate difference between herself and the ground.

    (D) What the astronaut notices depends on the direction of orbit relative to the rotation of the Earth.

2. Max kicks a football which then reaches a maximum height as seen below. Gloria who is watching from up a nearby tree, drops an identical football from the same height at the same time, as she observes Max's ball reaching maximum height. Note the bottom of the tree is level with where Max kicks the ball.



Which statement below best explains the motion of the two footballs?

(A) Max's ball hits the ground later.

(B) Max's ball hits the ground earlier.

(C) Both balls hit the ground at the same time.

(D) Both balls hit the ground at the same time only if the angle of launch is 90^0.

3. Which alternative in the table about the types of electrical discharge patterns formed in gas discharge tubes is correct?

	High pressure	*Lower pressure*	*Very low pressure*
(A)	Striations	Green glow on glass	Streamers
(B)	Streamers	Striations	Green glow on glass
(C)	Streamers	Green glow on glass	Striations
(D)	Green glow on glass	Striations	Streamers

4. Which statement about the Michelson-Morley experiment is correct?

(A) It was a valid experiment because it tested the principle of relativity.

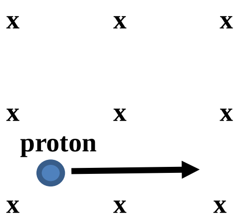
(B) It was a valid experiment because it took into account the known properties of light.

(C) It was an invalid experiment because it did not take into account the particle nature of light.

(D) It was an invalid experiment because the speed of Earth through the aether was not taken into account.

5. In which of the following does the proton in the magnetic field continue to move in a straight line?

(A)



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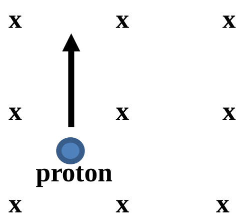
(B)



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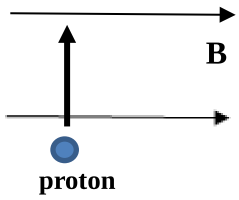
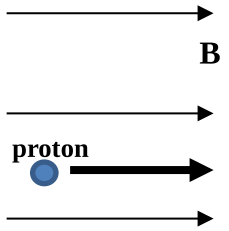
(C)



(D)



6. Which of the following is most correct for the electrons in atoms interacting with photons?

(A) Electrons that absorb photons will move closer to the nucleus and travel faster.

(B) Electrons will move to the next energy level if the photon is absorbed.

(C) Not all electrons will absorb photons.

(D) Electrons that absorb photons will move further away from the nucleus.

7. What was the technical development that led to the production and investigation of cathode rays?

(A) The ability to blow glass into thin tubes.

(B) The building of high voltage batteries.

(C) The use of photography to record results.

(D) The invention of good vacuum pumps.

8. What does the doping of a piece of pure silicon with a small amount of phosphorus do to the production of a semiconductor?

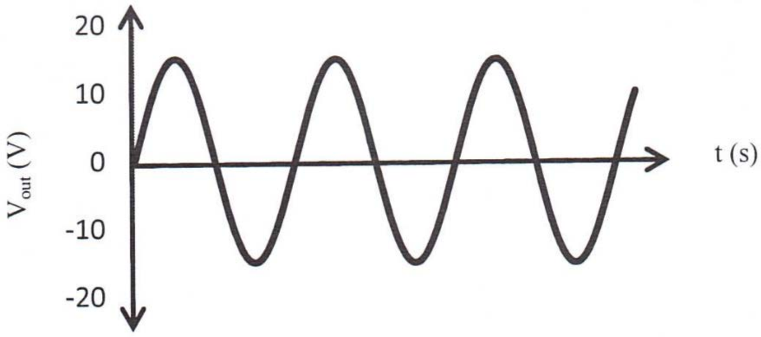
(A) Produces electrons as positive charge carriers

(B) Produces holes as positive charge carriers

(C) Produces electrons as negative charge carriers

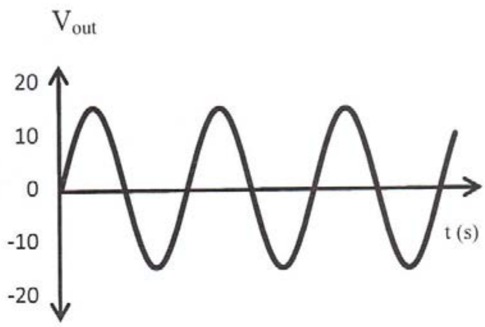
(D) Produces holes as negative charge carriers

9. A student moves a bar magnet at a constant rate in and out of a solenoid which is connected in a circuit to a voltmeter. The frequency at which this is done produces the following voltage output.

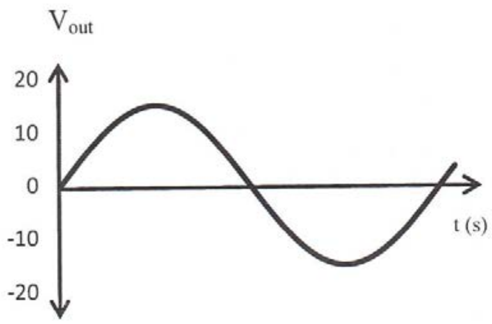


Which graph represents the most likely voltage output if the student reduces the frequency at which the magnet is moved in and out of the solenoid?

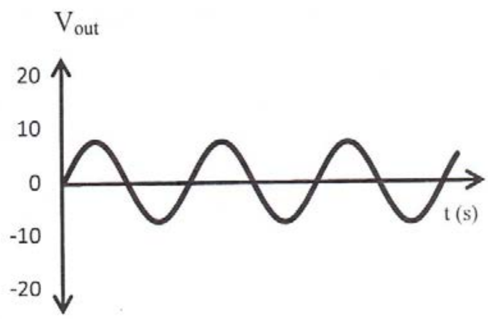
(A)



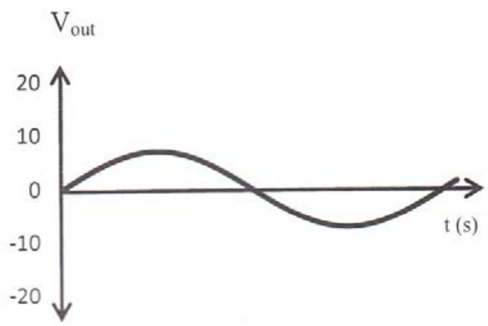
(B)



(C)



(D)



10. A transformer has a primary coil with 60 turns and a secondary coil with 2300 turns. If the primary voltage to the transformer is 110V, what is the secondary voltage?

(A) 2.4×10^{-4} V

(B) 2.4×10^{2} V

(C) 1.3×10^{3} V

(D) 4.2×10^{3} V

11. Choose the most correct statement below about an object in Earth orbit.

(A) If the object were to slow down, it would move into a lower orbit.

(B) An object cannot orbit the earth from north to south because of the Earth's rotation.

(C) All orbits will decay after millions of years.

(D) Objects in a stable orbit, no longer accelerate.

12. Which statement is most accurate about scientific information presented in the mass media compared with similar information presented in scientific journals?

(A) It is more accurate because the information is presented to an audience with a lower average reading level.

(B) It is less accurate because the purpose of presenting scientific information in the mass media is not well understood by publishers.

(C) It is more accurate because the audience of scientific journals is also part of the audience of mass media communication.

(D) It is less accurate because there are mixed purpose for presenting the information without the same mechanism for review of it.

13. A photon of green light has a wavelength of 500 nm. How much energy does this represent?

(A) 3.98×10^{-19} J

(B) 3.98×10^{-28} J

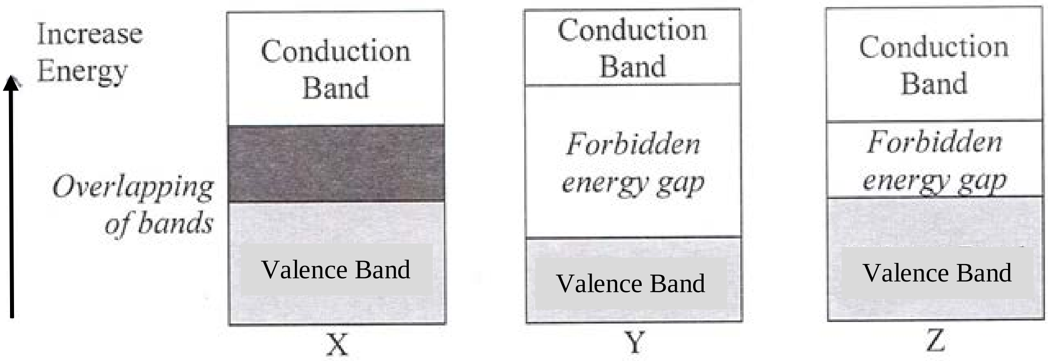
(C) 2.48×10^{-9} eV

(D) 2.48 J

14. A potential difference of 100V is applied between two identical, parallel aluminium plates which are separated by a distance of 10 mm.
In order to halve this electric field strength, which new arrangement should be used?

	Separation (mm)	*Potential difference* (V)	*Plates*
(A)	20	100	Aluminium
(B)	10	50	Perspex
(C)	5	100	Copper
(D)	20	50	Aluminium

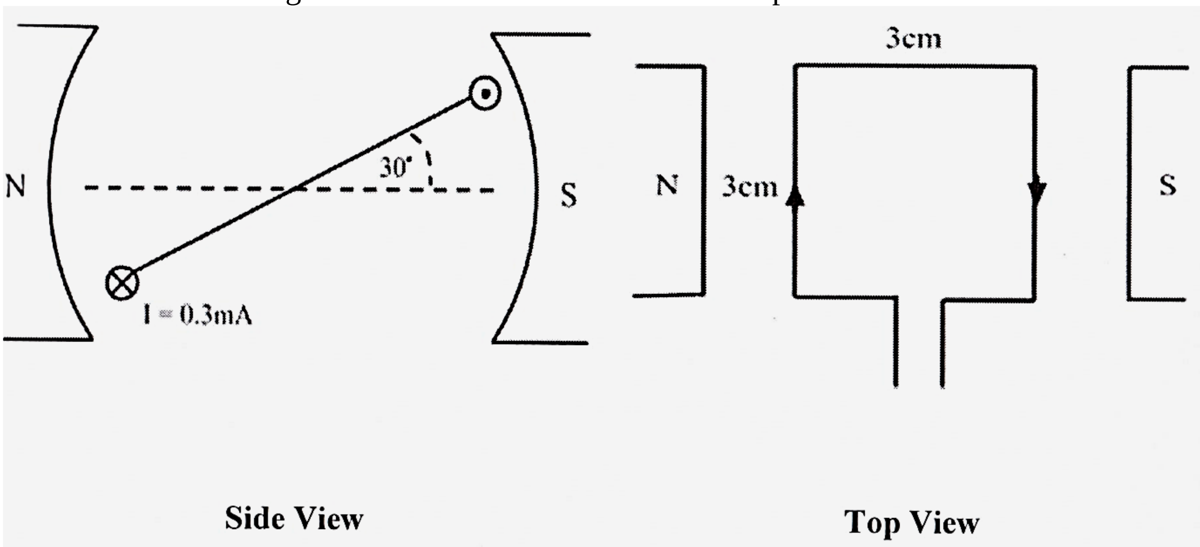
15. The following diagrams represents the electron band structures in different materials.



Which alternative correctly labels each material represented?

	X	Y	Z
(A)	Conductor	Insulator	Semiconductor
(B)	Insulator	Semiconductor	Conductor
(C)	Superconductor	Insulator	Conductor
(D)	Superconductor	Insulator	Semiconductor

16. The diagrams below show a single current-carrying loop of conducting wire with dimensions 3cm x 3cm in a circular magnetic field from a front view and a top view.



Using the front view to determine the direction of rotation, determine the torque on the loop if the strength of the magnetic field is **B** = 0.01T.

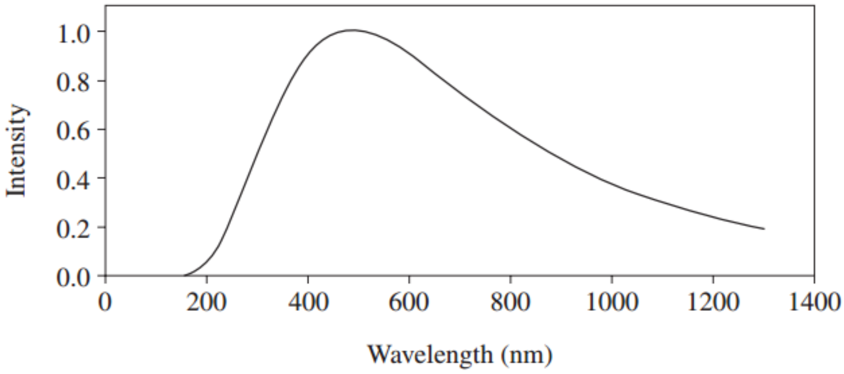
(A) 2.7×10^{-9} N.m anticlockwise

(B) 2.7×10^{-9} N.m clockwise

(C) 2.3×10^{-9} N.m anticlockwise

(D) 2.3×10^{-9} N.m clockwise

17. The graph shows the intensity–wavelength relationship of electromagnetic radiation emitted from a black body cavity.



In 1900, Planck proposed a mathematical formula that predicted an intensity–wavelength relationship consistent with the experimental data. The success of this formula depended on which of the following hypotheses?

(A) The intensity of light is dependent on the wavelength.

(B) Light is quantised, with the energy of light quanta depending on the frequency.

(C) Light is a wave whose intensity is readily expressed using mathematical formulae.

(D) Light is quantised, but not the energy.

18. The table shows the masses and radii of four imaginary planets expressed as ratios of the Earth's mass and radius.

Planet	**Mass** (x Earth's mass)	**Radius** (x Earth's radius)
W	4	3
X	2	4
Y	½	1
Z	½	½

On which of the four planets would an astronaut have the greatest weight?

(A) W

(B) X

(C) Y

(D) Z

19. A spacecraft begins a journey with a mass of 5×10^6 kg. On reaching a certain velocity the crew receive information that its mass is now 1×10^8 kg. The percentage of the speed of light that the ship is travelling at, is closest to?

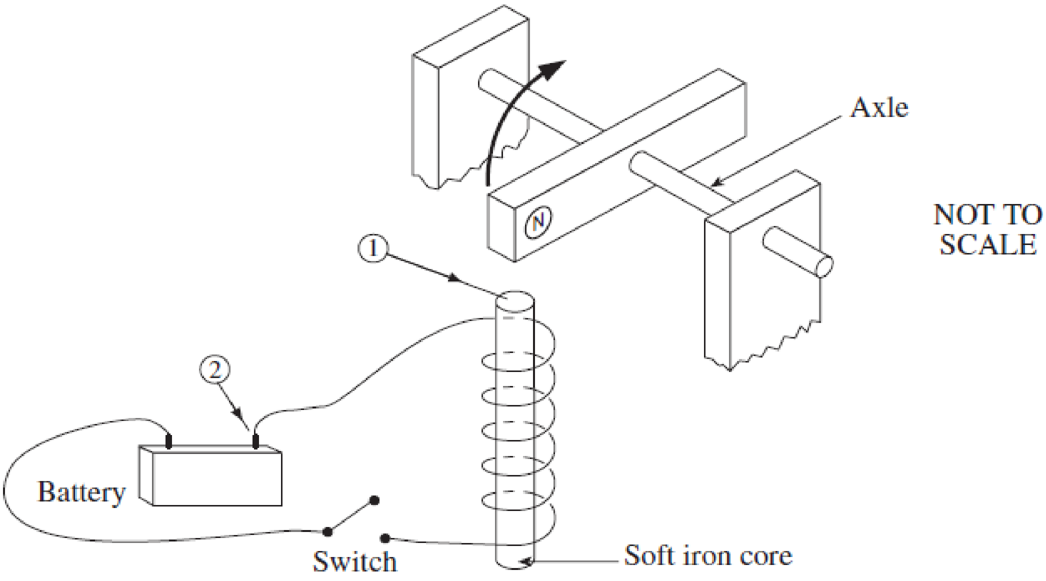
(A) 99.35

(B) 99.75

(C) 99.85

(D) 99.87

20. A magnet is free to spin. When the switch is closed, the magnet spins in the direction shown.



What is the magnetic polarity of the end of the soft iron core marked ① and the polarity of the battery terminal marked ②?

	Polarity of magnet ①	*Polarity of battery ②*
(A)	South	Positive
(B)	South	Negative
(C)	North	Positive
(D)	North	Negative

END OF MULTIPLE CHOICE QUESTIONS

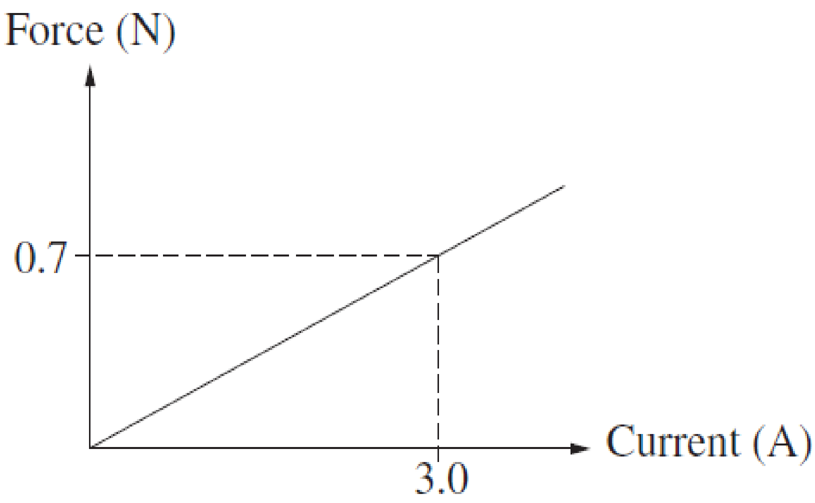
Fill in your student number

Student Number

Part B – 70 marks

Attempt all Questions on Space, Motors and Generators and Ideas to Implementation
Allow about 2 hours and 5 minutes for this section.
Answer the questions in the space provided.
Show all relevant working in questions involving calculations.

Question 21 (2 marks)
A student performed an experiment to measure the force on a long current-carrying conductor placed perpendicular to an external magnetic field.
The graph shows how the force on a 1.0 m length of the conductor varied as the current through the conductor changed.



Calculate the magnitude of the external magnetic field in this experiment?

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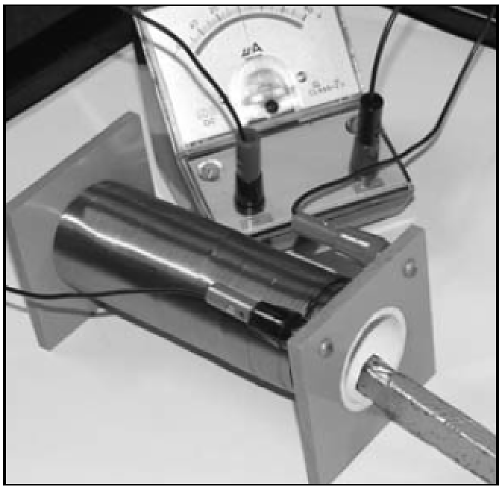
Question 22 (3 marks)

A student carried out a first-hand investigation into induction using equipment similar to that shown below.



Describe the effect on the generated electric current if

a) The strength of the magnet was doubled? **1**

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b) The magnet was placed inside the coil and held stationary? **1**

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c) The pole of the magnet inserted into the coil was reversed? **1**

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Question 23 (3 marks)

(a) Describe the meaning of the term work function in regard to the photoelectric effect. **1**

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(b) Orange light of wavelength 600 nm incident on a metal surface expels electrons with a maximum kinetic energy of 9.9×10^{-20} J. Calculate the work function of these electrons. **2**

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Question 24 (4 marks)

(a) A passenger of mass 60.0 kg is standing in a lift which is accelerating up at 1.00 m s^{-2}. **1**
Calculate the weight force of the passenger.

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(b) For the moons of Jupiter, Ganymede and Io:

Radius of orbit of Ganymede = 1.07 x 10^6 km

Period for one revolution of Ganymede = 7.15 days

Radius of revolution of Io = 4.22 x 10^5 km

Calculate the period of revolution of Io about the planet Jupiter (in days). **3**

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Question 25 (4 marks)

(a) Ben, the spaceship's co-pilot, tells the Captain that the escape velocity of the Earth is 11 kms^{-1}. Therefore, the spaceship must obtain this speed in order to commence the ships journey to Mars by escaping the Earth. Should the Captain be happy with the co-pilot's advice? Explain. **2**

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Question 25 continues on the next page

(b) The gravitational force between two protons is 4.65 x 10^{-37} N. Calculate their separation distance, correct to 2 significant figures? **2**

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Question 26 (3 marks)
Describe an investigation or experiment you studied that would demonstrate the production and reception of radio waves.

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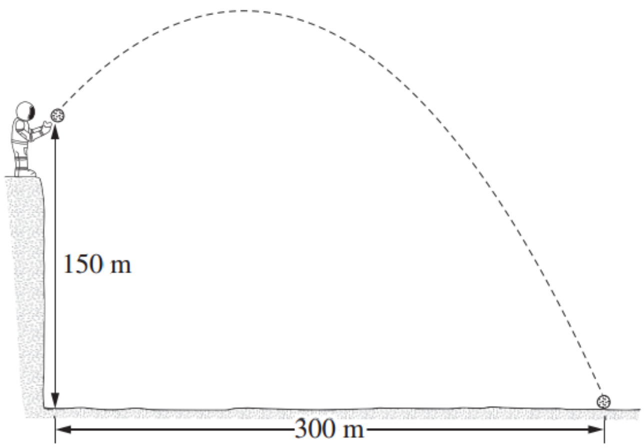
Question 27 starts on the next page

Question 27 (5 marks)
An astronaut on the Moon throws a stone from the top of a cliff. The stone hits the ground below 21.0 seconds later. The acceleration due to gravity on the moon is 1.6 ms^{-2}.



(a) Calculate the horizontal component of the stone's initial velocity. Show your working. **1**

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(b) Calculate the vertical component of the stone's initial velocity. Show your working. **2**

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(c) On the diagram, sketch the path that the stone would follow if the acceleration due to gravity was higher. The initial velocity is the same. **2**

Question 28 (8 marks)
(a) Calculate the voltage between two parallel plates which are 1 mm apart, if the electric field intensity is 2.4×10^{-6} NC^{-1} between the plates. **2**

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Question 28 continues on the next page

(b) Determine a formula for the force on a charge, q, between two parallel plates. Explain why this force is constant regardless of the position of the charge? **3**

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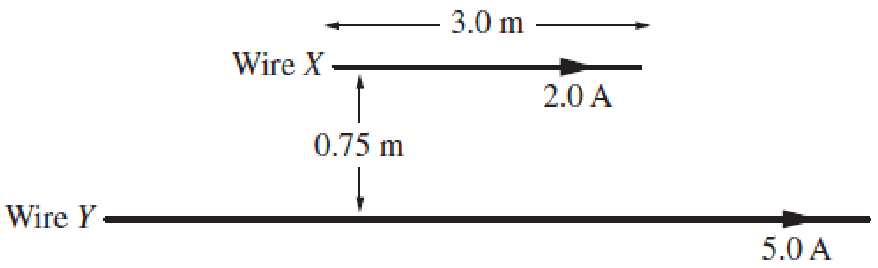
(c) Two parallel wires are separated by a distance of 0.75m. Wire X is 3.0m long and carries a current of 2.0A. Wire Y can be considered to be indefinitely long and carries a current of 5.0A. Both currents flow in the same direction along the wires.



i. Determine the direction of the force that exists between the two wires. **1**

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ii. Determine the magnitude of the force that exists between the two wires. **2**

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Question 29 (4 marks)

An observer, Owen, in a manned space vehicle which is swooping low over the Earth's surface at a speed of 0.8 c, sees two simultaneous explosions below him at points A and B. At this instant he is just over a point C, halfway between A and B. Mary Ellen is at rest at B.

(a) Will Mary Ellen agree with Owen that the explosions were simultaneous? Discuss. **2**

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(b) Mary Ellen is now sitting at a railway station whose platform is 100 m long. Owen is now in a train which, at rest, is 150 m long.

Determine the speed the train must be travelling so that Mary Ellen observes the whole length of the train to fit exactly along the platform? **2**

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Question 30 (4 marks)

A person on the Earth's surface will actually weigh a little less because of the Earth's rotation.

(a) Explain briefly why this is so? **1**

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(b) Calculate the length of the day in order for people to become weightless on the equator due to an increase in the Earth's rotational speed? (Radius of the Earth 6378 km) **3**

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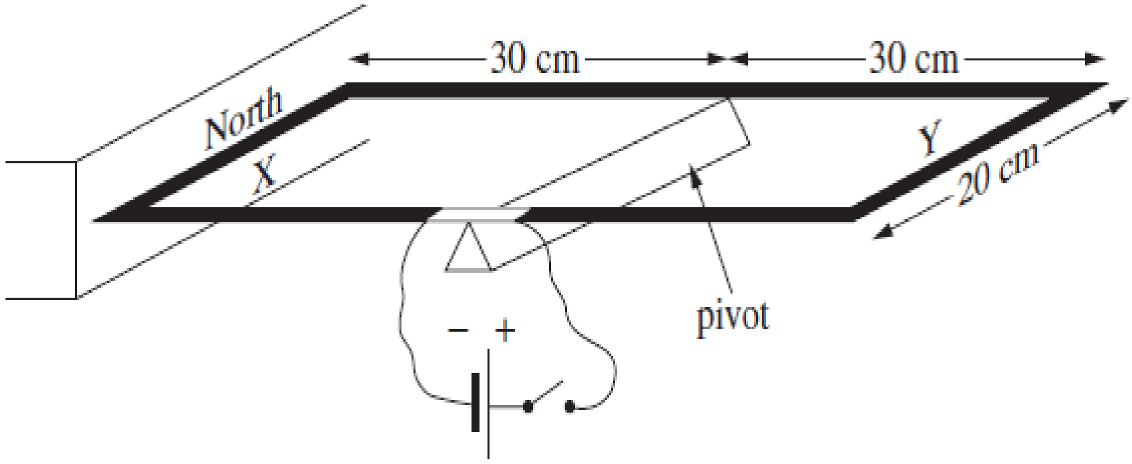
Question 31 (5 marks)

A rectangular wire loop is connected to a DC power supply. Side X of the loop is placed next to a magnet. The loop is free to rotate about a pivot.



A mass of 60g is attached to side X of the loop to prevent rotation and a current of 30 A is supplied to the loop.

a) Calculate the torque provided by the 60g mass. **2**

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b) Calculate the magnetic field strength around side X. **3**

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Question 32 (5 marks)

(a) The spiralling re-entry of a LEO satellite through Earth's atmosphere would destroy it. Space Shuttle missions travel in LEO orbits but their re-entry is controlled.

Describe how destruction of the Space Shuttle is prevented during its re-entry. **3**

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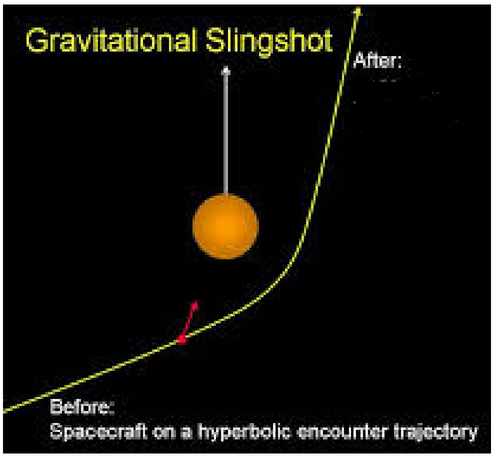
(b) By referring to the diagram below, explain the purpose of a gravitational slingshot manoeuvre and its effect on the spacecraft and planet's velocity. **2**



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Question 33 (8 marks) **Marks**

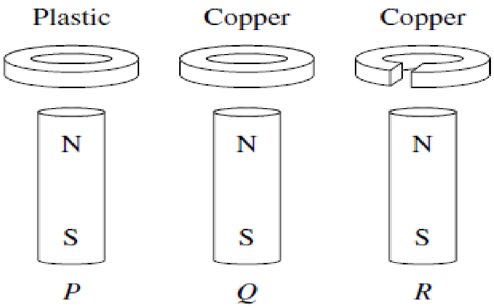
(a) In a simple DC motor, describe the role of

i. The commutator **1**

ii. The stator **1**

(b) Most electric motors today are ‘AC induction motors’. Explain why this is so by giving two advantages and one disadvantage of ‘AC induction motors’. **3**

(c) Three rings are dropped at the same time over identical magnets as shown below. **3**



Explain the order in which the rings will reach the bottom of the magnets.

Question 34 (6 marks)

(a) Explain what the effect of 'doping' has on the band structure of silicon atoms. **2**

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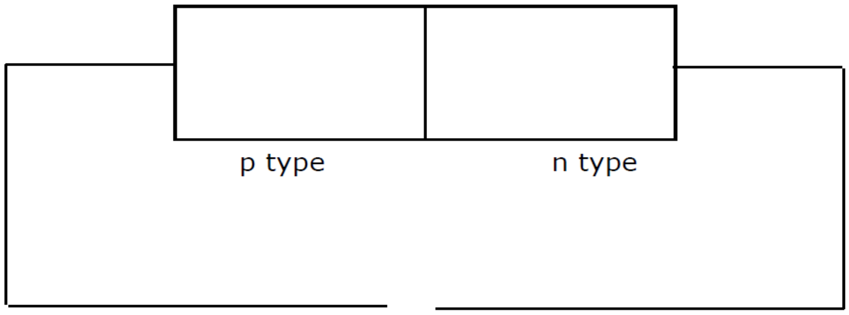
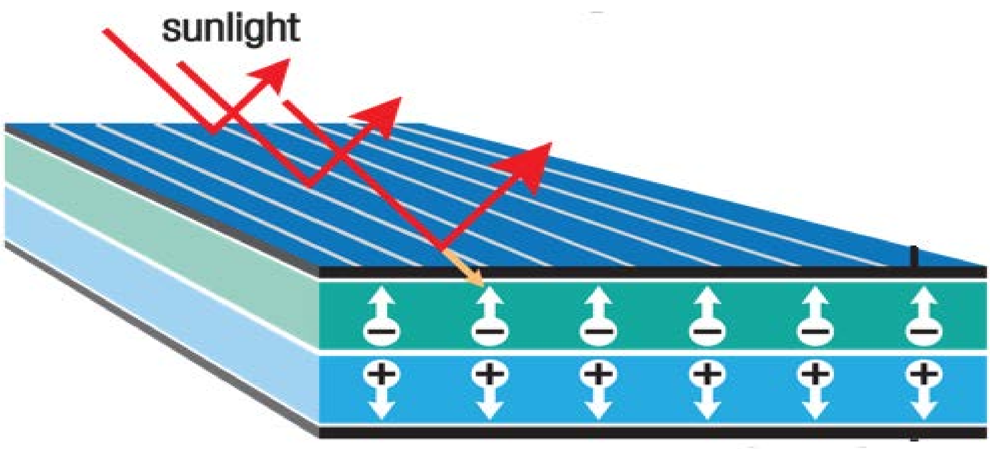
(b) Add 2 sets of positive + and negative - charges to indicate **2**

- the movement of electric charge carriers
- the battery terminals

when a p-n junction is connected to a battery, as in the diagram.



(c) The diagram below shows sunlight incident on a cross section of a solar cell. By first completing the diagram by inserting a circuit with a galvanometer and showing the conventional current direction, briefly explain how the solar cell would operate. **2**



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Question 35 (6 marks)

Historically, scientists have made a number of observations in their photoelectric effect experiments. Some of their observations were unable to be explained by existing theory.

Outline these observations and how Einstein was able to explain each one using Planck's work.

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END OF QUESTIONS FOR PART B

2014 Trial Higher School Certificate Physics

Student Number

Section II

Attempt all parts of question 36.
Allow about 20 minutes for this part
<u>Answer the question in the spaces provided</u>.
Show all relevant working in questions involving calculations.

Marks

Question 36 Quanta to Quarks (10 marks)

(a) The diagram represents the four spectral lines in the visible region of the hydrogen spectrum known as the Balmer series.

(i) Explain how other spectral lines can exist but are not visible to the human eye. **1**

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(ii) Determine the wavelength of the radiation produced when an electron in the hydrogen atom moves from the fifth energy level to the third energy level. **2**

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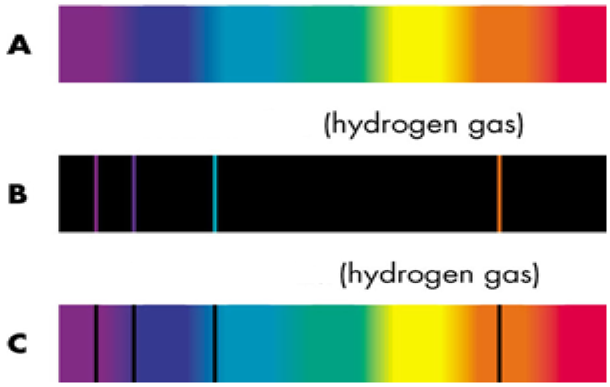
(b) Explain the difference between absorption and emission spectra by referring to the diagram, and ensuring you distinguish between each of the three figures A, B and C. **3**



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(c) Discuss the development of theories of the structure of the atom with reference to the work of any TWO of the following: Thomson, Rutherford, Bohr.
In each case, indicate one piece of experimental evidence which supported the idea. **4**

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End of Questions on Quanta to Quarks

End of Examination